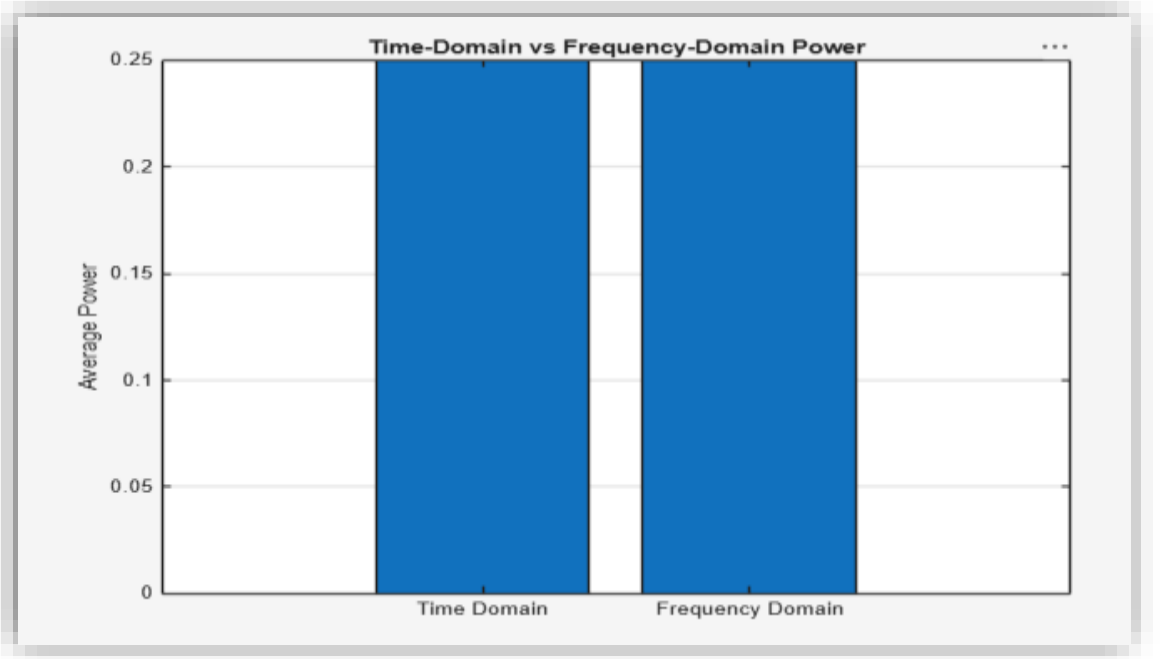
Experiment – 6
Parseval's Relation and Harmonic Spectrum Analysis

(iv)

/MATLAB Drive/Experiment_6_4.m

```
clc;
clear;
close all;
%% 6(iv) Comparison of Time-Domain Power with Frequency-Domain Harmonic Power
figure;
% Time-Domain and Frequency-Domain Power Comparison
f0 = 1; T = 1/f0;
t = linspace(0,T,4000);
x = max(sin(2*pi*f0*t),0);
P_time = (1/T)*trapz(t,x.^2);
N = 20;
a0 = (2/T)*trapz(t,x);
an = zeros(1,N); bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end
P_dc = a0^2/4;
P_harmonics = 0.5*sum(an.^2+bn.^2);
P_frequency = P_dc + P_harmonics;
fprintf('Time-domain power = %.6f\n',P_time);
fprintf('DC power = %.6f\n',P_dc);
fprintf('Harmonic power = %.6f\n',P_harmonics);
fprintf('Frequency-domain power = %.6f\n',P_frequency);
bar([P_time P_frequency]);
set(gca,'XTickLabel',{'Time Domain','Frequency Domain'});
ylabel('Average Power'); grid on;
title('Time-Domain vs Frequency-Domain Power');
```

Code



Output